



Product Optimization & Revenue Contribution Analysis

for Afficionado Coffee Roasters

Research Paper — EDA, Insights & Recommendations

Dataset: Transaction-level sales data, FY 2025

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1. Introduction & Background

Retail success in a specialty food and beverage business depends on understanding three core questions: which products customers prefer, which products generate the most revenue, and how revenue is distributed across the menu. Without structured analytics, decisions about which items to promote, simplify, or discontinue are often made on intuition rather than evidence — a pattern that risks two costly outcomes: keeping high-priced products with low sales frequency on the menu, and running an overcrowded menu that slows service and confuses customers.

Afficionado Coffee Roasters records transaction-level sales data across its stores, but historically has lacked clear visibility into product popularity versus profitability, and into how dependent the business is on a small number of core categories. This project addresses that gap.

1.1 Problem Statement

Although transaction-level product data is available, Afficionado Coffee Roasters lacks clear visibility into product popularity versus profitability, and category-level revenue dependency insights. Without structured product analytics, menu decisions are often intuition-driven, leading to inefficiencies and missed revenue opportunities.

1.2 Objectives

Primary objectives:

- Identify top-selling and least-selling products
- Quantify revenue contribution by product and category
- Measure revenue concentration across the menu

Secondary objectives:

- Support menu simplification and optimization
- Identify high-impact "hero" products
- Highlight low-performing products for review or redesign

2. Dataset Description

The dataset consists of 149,116 individual sales transactions recorded across FY 2025, with 11 fields per transaction. Data validation confirmed the dataset is complete and clean: zero missing values across all columns, zero duplicate transaction IDs, and zero records with non-positive quantity or price.

Column	Description
transaction_id	Unique identifier per transaction
year	Transaction year (2025)
transaction_time	Time of transaction (HH:MM:SS)
transaction_qty	Quantity purchased
unit_price	Price per unit
store_id	Store identifier
store_location	Physical store location
product_id	Unique identifier for each product
product_category	Broad product group (e.g. Coffee, Tea)
product_type	Product variant within the category
product_detail	Detailed attributes such as flavor or blend

Dataset scale at a glance:

- 149,116 transactions
- 29 distinct product types across 9 product categories
- Total revenue: ₹698,812.33

3. Analytical Methodology

The analysis followed six sequential stages:

- **Data Ingestion & Validation** — loading transaction-level data and confirming there were no missing values, duplicate transactions, or invalid (zero/negative) quantities or prices.
- **Revenue Computation** — Revenue = transaction_qty × unit_price, computed per transaction and aggregated by product, product type, and product category.
- **Product Popularity Analysis** — total units sold per product type, ranked to identify top and bottom performers by volume.
- **Revenue Contribution Analysis** — total revenue and revenue share (%) per product type, with a comparison between volume rank and revenue rank to surface mismatches.
- **Category & Product-Type Performance** — revenue share by category, and product-type contribution within each category, to assess dependency on core categories.
- **Revenue Concentration & Menu Balance** — Pareto (80/20) analysis to identify revenue-anchor products versus long-tail products, and to evaluate menu diversification risk.

4. EDA & Findings

4.1 Product Popularity — Top Products by Sales Volume

Ranked by total units sold (top 10 of 29 product types):

Rank	Product Type	Units Sold
1	Brewed Chai tea	26,250
2	Gourmet brewed coffee	25,973
3	Barista Espresso	24,943
4	Brewed Black tea	17,462
5	Hot chocolate	17,457
6	Brewed herbal tea	17,328
7	Organic brewed coffee	13,012
8	Drip coffee	12,891
9	Premium brewed coffee	12,431
10	Scone	10,465

4.2 Revenue Contribution — Top Products by Revenue

Ranked by total revenue generated (top 10 of 29 product types):

Rank	Product Type	Revenue (₹)	% of Total Revenue
1	Barista Espresso	91,406.20	13.08%
2	Brewed Chai tea	77,081.95	11.03%
3	Hot chocolate	72,416.00	10.36%
4	Gourmet brewed coffee	70,034.60	10.02%
5	Brewed Black tea	47,932.00	6.86%

Rank	Product Type	Revenue (₹)	% of Total Revenue
6	Brewed herbal tea	47,539.50	6.80%
7	Premium brewed coffee	38,781.15	5.55%
8	Organic brewed coffee	37,746.50	5.40%
9	Scone	36,866.12	5.28%
10	Drip coffee	31,984.00	4.58%

Notably, the two rankings diverge: Brewed Chai tea sells the most units but ranks second in revenue, while Barista Espresso sells fewer units (3rd by volume) but generates the most revenue — reflecting its higher per-unit price. This gap between volume rank and revenue rank was quantified for every product type (Section 4.3).

4.3 Volume Rank vs. Revenue Rank — Premium and Underperforming Items

Comparing each product's sales-volume rank against its revenue rank highlights two distinct groups. Products whose revenue rank is meaningfully better than their volume rank are premium, high-value items — they don't need to sell in bulk to contribute strongly to revenue. Products whose revenue rank is worse than their volume rank sell in reasonable quantity but contribute disproportionately little revenue, making them candidates for repricing or review.

Product Type	Volume Rank	Revenue Rank	Interpretation
Barista Espresso	3	1	Premium / high-value
Clothing	18	25	Sells reasonably, low revenue contribution
House blend Beans	27	22	Sells reasonably, low revenue contribution
Premium Beans	20	15	Sells reasonably, low revenue contribution
Sugar free syrup	15	26	Underperforms relative to volume
Regular syrup	12	19	Underperforms relative to volume

4.4 Category & Product-Type Performance

Revenue share by product category:

Category	Revenue (₹)	% of Total Revenue
Coffee	269,952.45	38.63%
Tea	196,405.95	28.11%
Bakery	82,315.64	11.78%
Drinking Chocolate	72,416.00	10.36%
Coffee beans	40,085.25	5.74%
Branded	13,607.00	1.95%
Loose Tea	11,213.60	1.60%
Flavours	8,408.80	1.20%
Packaged Chocolate	4,407.64	0.63%

Coffee and Tea together account for 66.7% of total revenue. This concentration indicates the business is meaningfully dependent on these two categories — a strategic strength (clear core identity) but also a risk if demand for either category shifts.

4.5 Revenue Concentration — Pareto (80/20) Analysis

Ranking all 29 product types by revenue and examining cumulative contribution shows that a small subset of the menu drives the large majority of revenue:

Metric	Value
Product types generating ~80% of revenue	10 of 29 (34.5%)
Revenue share of top 10 product types	78.96%
Revenue share of remaining 19 product types (long tail)	~21%

This is the central finding of the analysis: roughly one-third of the product line (the 10 revenue-anchor products — led by Barista Espresso, Brewed Chai tea, Hot chocolate, and Gourmet brewed coffee) generates nearly four-fifths of total revenue. The remaining ~19 product types form a long tail that contributes only about a fifth of revenue between them, despite adding menu complexity, inventory overhead, and operational load.

5. Key Performance Indicators (KPIs)

KPI	Description	Key Result
Product Revenue Contribution (%)	Share of total revenue	Barista Espresso leads at 13.08%
Product Sales Volume	Popularity indicator	Brewed Chai tea leads at 26,250 units
Category Revenue Share	Business dependency	Coffee 38.63%, Tea 28.11% (66.7% combined)
Revenue Concentration Ratio	Menu risk exposure	78.96% of revenue from top 10 of 29 product types
Product Efficiency Score	Revenue per unit sold	Premium Beans leads at ₹35.92/unit

The Product Efficiency Score (revenue ÷ units sold) surfaces a different view from raw revenue: items like Premium Beans, Clothing, and Organic Beans generate the most revenue per unit sold, even though their total volumes are low. These are high-margin, low-volume specialty items rather than everyday high-traffic items such as Brewed Chai tea, which has a much lower efficiency score (₹9.71/unit) despite topping the volume chart.

6. Recommendations

- Protect and promote the top 10 revenue-anchor products (led by Barista Espresso, Brewed Chai tea, Hot chocolate, and Gourmet brewed coffee) — they drive ~79% of revenue and should anchor menu design, staffing, and inventory priorities.
- Review the long-tail products (the ~19 product types contributing only ~21% of revenue combined) for potential simplification — either through targeted promotion to raise their contribution, repricing, or discontinuation of the weakest performers.
- Investigate underperforming-relative-to-volume items (e.g. Sugar free syrup, Regular syrup) — these sell in moderate volume but contribute disproportionately little revenue, suggesting a pricing or margin issue worth a closer look.
- Leverage high-efficiency niche items (e.g. Premium Beans, Organic Beans) as premium/upsell offerings, since they generate strong revenue per unit despite low volume.
- Monitor category concentration risk — with Coffee and Tea contributing two-thirds of revenue, the business should track early signals of demand shifts in these categories and consider gradual diversification.

7. Conclusion

This project delivers actionable product intelligence by shifting focus from when customers buy to what they buy and what drives revenue. By uncovering product-level performance and revenue concentration patterns, Afficionado Coffee Roasters can optimize its menu, improve operational efficiency, and strengthen long-term profitability. The accompanying Streamlit dashboard makes these findings explorable on an ongoing basis, with filters for category, store location, and top-N product views.

Appendix: Supporting Visuals

The following charts, generated during the analysis, support the findings above and can be inserted here: (1) Top 10 Products by Revenue — bar chart; (2) Revenue Share by Category — pie chart; (3) Popularity vs. Revenue — scatter plot; (4) Pareto Analysis — cumulative revenue concentration chart.